Performance Evaluation of Routing Protocols in Software-Defined Networks (SDN)

A CAPSTONE PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Performance Evaluation of Routing Protocols in Software-Defined Networks (SDN)" has been carried out by **K.Mahesh (192312376)** under the supervision of **Dr. V. Parthipan** and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **K.Mahesh (192312376)** of the ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING department, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "Performance Evaluation of Routing Protocols in Software-Defined Networks (SDN)" , representing my individual contribution to this team project — is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Signature of the Students with Names

ABSTRACT

Software-Defined Networking (SDN) is a modern networking approach that separates the control plane from the data plane, enabling centralized network management and improved flexibility. This project, Performance Evaluation of Routing Protocols in Software-Defined Networks (SDN), focuses on designing, implementing, and analysing an SDN-based network model using network simulation tools. The primary objective is to evaluate how SDN improves routing efficiency, traffic management, and overall network performance compared to traditional networking methods.

The proposed system includes an SDN Controller that acts as the central decision-making unit, managing all network devices and controlling packet forwarding through flow rules. Flow management is implemented to organise data transmission efficiently by identifying packet flows based on parameters such as source IP address, destination IP address, and communication protocol. A virtual network topology is created to simulate different network conditions and evaluate routing behaviour, traffic handling, and resource utilisation.

The simulation results demonstrate successful communication between network devices with reduced congestion, improved traffic control, and better resource allocation. Centralised management simplifies network configuration, minimises manual intervention, and allows the network to adapt quickly to changing requirements. The proposed model also provides enhanced scalability, flexibility, and efficient utilisation of network resources while reducing the cost of testing through simulation.

Overall, this project proves that Software-Defined Networking offers a reliable and efficient solution for modern communication networks. The performance evaluation highlights the advantages of SDN in achieving improved routing efficiency, simplified management, and optimised network performance, making it a promising technology for future enterprise, cloud, and data centre networking applications.

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CHAPTER 1 INTRODUCTION

Computer networks form the backbone of modern digital communication, enabling data exchange between devices across the globe. As the scale and complexity of networks continue to grow, traditional networking architectures — where control logic is distributed across individual routers and switches — face increasing difficulty in adapting to dynamic traffic patterns, scaling efficiently, and simplifying management. Software-Defined Networking (SDN) has emerged as a transformative paradigm that addresses these limitations by decoupling the control plane from the data plane, allowing centralized, programmable control over the entire network.

This project, titled "Performance Evaluation of Routing Protocols in Software-Defined Networks (SDN)," investigates how SDN-based routing compares with conventional routing approaches in terms of throughput, latency, packet loss, and adaptability to changing network conditions. Using an SDN controller as the centralized decision-making unit, network devices are managed through flow rules rather than distributed routing tables, allowing more flexible and efficient control over packet forwarding.

Traditionally, routing decisions were made independently by each router using protocols such as RIP, OSPF, or static routing tables, which often led to slower convergence, inconsistent traffic engineering, and limited visibility into overall network state. The SDN-based model implemented in this project eliminates these issues by providing a centralized platform where routing decisions, flow management, and traffic monitoring can be observed and controlled from a single point, using tools such as a virtual network topology, an OpenFlow-based controller, and traffic simulation utilities.

The system developed for this project consists of key components including the SDN Controller, Network Topology Simulation, Flow and Routing Management, and Performance Monitoring. The controller acts as the brain of the network, installing flow rules on switches based on the routing protocol under evaluation. The topology simulation creates a virtual network of hosts, switches, and links to emulate realistic traffic conditions. The flow and routing management module identifies and forwards packet flows based on parameters such as source and destination

IP addresses and protocol type. The performance monitoring module collects and analyses metrics such as throughput, delay, jitter, and packet loss for each routing protocol under test.

The primary objective of this project is to evaluate how SDN-based centralized routing improves network performance, simplifies configuration, and adapts to varying traffic loads compared to traditional distributed routing methods. By integrating network simulation tools, controller programming, and performance analysis techniques, this project demonstrates the practical benefits of Software-Defined Networking and contributes to a deeper understanding of how modern networks can be designed to be more efficient, scalable, and manageable in today's increasingly data-driven digital environment.

CHAPTER 2 PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Traditional computer networks rely on distributed routing protocols in which each router independently computes and maintains its own routing table based on local information exchanged with neighbouring devices. While this approach has served networks well for decades, it presents significant limitations as network size, traffic volume, and application demands continue to increase. Distributed control makes it difficult to obtain a consolidated, real-time view of the entire network, which in turn complicates traffic engineering, congestion management, and rapid fault recovery.

In many enterprise and data centre environments, network administrators must configure routing policies individually on each device, which is time-consuming, error-prone, and difficult to scale. Static or semi-static routing configurations often struggle to adapt quickly to changing traffic patterns, link failures, or new application requirements, leading to inefficient use of network resources, unpredictable latency, and reduced overall performance.

2.2 Evidence of the Problem

Industry reports and academic research consistently highlight those distributed, hardware-bound routing architectures are increasingly unable to keep pace with the demands of cloud computing, virtualization, and large-scale data centre traffic. Studies on conventional routing protocols such as RIP and OSPF show that convergence after topology changes can take several seconds, during which packets may be dropped or misrouted, degrading the experience for latency-sensitive applications such as video conferencing and real-time analytics.

Furthermore, network operators frequently report difficulties in enforcing consistent traffic engineering policies across heterogeneous vendor equipment, since each device implements routing logic independently. This lack of centralized visibility and control makes it challenging to optimize path selection, balance load across links, or quickly diagnose the root cause of performance degradation.

2.3 Stakeholders

Several stakeholders are directly affected by the performance and manageability of routing infrastructure within a network.

Network Administrators

Administrators are responsible for configuring, monitoring, and troubleshooting routing behaviour across the network, and they benefit directly from centralized, programmable control.

Cloud and Data Centre Operators

Operators rely on efficient, adaptable routing to maintain service-level agreements, balance load across servers, and minimize downtime.

End Users and Applications

End users expect consistent, low-latency connectivity, particularly for real-time applications such as video streaming, gaming, and voice communication.

Internet Service Providers (ISPs)

ISPs depend on scalable and efficient routing to manage large volumes of traffic while controlling operational costs.

Researchers and Developers

Researchers use SDN platforms to design, test, and evaluate new routing algorithms and traffic engineering strategies in a controlled, reproducible environment.

2.4 Supporting Data and Research

Research conducted by organizations such as the Open Networking Foundation (ONF) demonstrates that Software-Defined Networking, through protocols such as OpenFlow, enables centralized control planes that can significantly reduce configuration complexity and improve network responsiveness. Studies comparing SDN-based routing with traditional distributed protocols report measurable improvements in convergence time, load balancing efficiency, and fault recovery speed when routing decisions are made by a centralized controller with a global view of the network.

Simulation-based studies using tools such as Mininet and OpenFlow-compatible controllers show that centralized routing can dynamically recompute optimal paths in response to link failures or congestion far more quickly than distributed protocols relying on periodic update exchanges. These findings support the need for a comprehensive evaluation of SDN-based routing protocols, comparing their throughput, latency, and adaptability against conventional routing methods to quantify the practical benefits of centralized network control.

CHAPTER 3 SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The development of the SDN-based routing evaluation system followed a structured and systematic approach. Initial requirements were gathered by analysing the limitations of traditional distributed routing protocols, including slow convergence, limited visibility, and difficulty in traffic engineering. Based on these requirements, a virtual network environment was designed to emulate realistic topologies consisting of multiple hosts, switches, and links, over which different routing strategies could be implemented and tested under a common SDN controller.

The system architecture was divided into key modules: Network Topology Simulation, SDN Controller and Flow Management, Routing Protocol Implementation, and Performance Monitoring and Analysis. Each module was developed and tested independently before being integrated into a complete evaluation pipeline. The controller was configured to install and update flow entries on virtual switches dynamically, allowing routing behaviour to be modified and observed without altering the physical or virtual topology.

Continuous testing was carried out throughout development to validate that packets were being forwarded correctly according to the active routing policy, and that performance metrics were being captured accurately for each test scenario.

3.2 Tools and Technologies Used

The project was implemented using a combination of network simulation and controller technologies suited to SDN research and evaluation.

Network Emulation

- Mininet — for creating virtual network topologies of hosts, switches, and links

SDN Control Plane

- OpenFlow protocol for controller-to-switch communication
- An OpenFlow-compatible SDN controller (e.g., Ryu / POX / ONOS) for centralized flow management

Analysis and Scripting

- Python for controller logic, automation scripts, and data processing
- Wireshark for packet capture and traffic inspection
- iPerf for throughput and bandwidth measurement

These tools collectively support the emulation of realistic network conditions, centralized routing control, and accurate measurement of performance metrics across different routing strategies.

3.3 Solution Overview

The proposed solution establishes a centralized SDN Controller that maintains a global view of the virtual network topology and manages packet forwarding through flow rules installed on OpenFlow-enabled switches. Multiple routing strategies — including a shortest-path (Dijkstra-based) approach and a traditional distributed-style protocol emulated for comparison — are implemented within the controller logic, allowing each to be evaluated under identical network and traffic conditions.

When traffic is generated between hosts in the emulated topology, the controller identifies new flows based on parameters such as source IP address, destination IP address, and protocol type, then computes and installs the appropriate forwarding path across the relevant switches. As traffic patterns or link conditions change, the controller can recompute routes and update flow tables dynamically, demonstrating the adaptability of centralized routing control.

Performance data — including throughput, end-to-end latency, jitter, and packet loss — is collected for each routing strategy under varying traffic loads and topology conditions, and the results are compiled into comparative reports and graphs to highlight the relative strengths of SDN-based routing over traditional approaches.

3.4 Engineering Standards Applied

The project follows established networking and software engineering standards to ensure the reliability, reproducibility, and quality of the evaluation.

Networking Standards

- OpenFlow protocol specification for controller-switch communication
- Standard IP addressing and subnetting practices
- Consistent topology documentation and version control

Software Engineering Standards

- Modular controller and script design for maintainability
- Repeatable, scripted experiment setup for reproducible results
- Proper documentation of configurations and test scenarios

Measurement Standards

- Consistent traffic generation parameters across all test runs
- Standardized metrics: throughput, latency, jitter, packet loss

Adhering to these standards ensures that performance comparisons between routing protocols are fair, consistent, and reproducible across multiple test runs.

3.5 Solution Justification

The SDN-based evaluation framework provides an effective and controlled environment for comparing routing strategies that would otherwise be difficult to test consistently on physical hardware. Traditional distributed routing protocols require per-device configuration and offer limited insight into real-time network state, making systematic performance comparison challenging. The centralized controller used in this project simplifies experimentation by allowing routing logic to be modified and re-tested from a single point of control.

The use of network emulation tools such as Mininet allows realistic topologies to be created and reset quickly, reducing the cost and complexity of testing compared to physical lab setups. The integration of packet capture and throughput measurement tools ensures that performance results are grounded in objective, measurable data rather than theoretical assumptions, giving the project's conclusions practical credibility.

CHAPTER 4 RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The implementation of the SDN-based routing evaluation framework successfully achieved its primary objective of comparing centralized SDN routing against traditional distributed routing approaches. The emulated topology allowed multiple routing strategies to be tested under identical conditions, providing a fair basis for performance comparison.

The SDN Controller Module effectively computed and installed forwarding paths with lower convergence delay following simulated link changes compared to the emulated distributed protocol. The Performance Monitoring Module captured consistent measurements of throughput, latency, jitter, and packet loss across all test scenarios, showing that SDN-based routing generally achieved higher throughput and lower average latency, particularly under conditions involving topology changes or fluctuating traffic load.

Overall, the results confirmed that centralized, programmable routing control improves network responsiveness, simplifies path optimization, and offers better adaptability to dynamic conditions than traditional distributed routing, while also making it significantly easier to monitor and analyse network-wide behaviour from a single control point.

4.2 Challenges Encountered

Several challenges arose during the development and testing of the system. One major challenge was designing a virtual topology complex enough to reveal meaningful performance differences between routing strategies while still remaining manageable to configure and test repeatedly.

Another challenge involved ensuring accurate and consistent measurement of performance metrics, since background emulation overhead in the virtual environment could introduce measurement noise. Careful test design, repeated trials, and averaging of results were used to reduce the impact of this variability.

Implementing dynamic route recomputation within the controller logic also required careful handling of flow table updates to avoid transient loops or dropped packets during transitions, which was addressed through incremental testing and controller logic refinement.

4.3 Possible Improvements

Although the current evaluation framework performs effectively, several enhancements could further improve its depth and applicability.

Multi-Controller Architecture

Extending the framework to support distributed or hierarchical SDN controllers would allow evaluation of scalability and fault tolerance in larger networks.

QoS-Aware Routing

Incorporating Quality-of-Service parameters such as bandwidth reservation and priority queuing would enable evaluation of routing under application-specific performance requirements.

AI-Based Traffic Prediction

Machine learning models could be integrated to predict traffic patterns and proactively adjust routing decisions before congestion occurs.

Security-Aware Routing

Future versions could evaluate routing strategies that incorporate security considerations, such as avoiding compromised or untrusted network segments.

Larger-Scale Emulation

Testing on larger, more complex topologies would provide stronger evidence of how SDN-based routing scales in enterprise and data centre environments.

4.4 Recommendations

For future development, it is recommended that the evaluation framework be extended to support multiple SDN controllers and larger, more realistic network topologies to better represent enterprise and data centre conditions. Routing algorithms should be benchmarked against additional metrics such as energy efficiency and controller processing overhead to provide a more complete performance picture.

The adoption of QoS-aware and AI-assisted routing strategies is also recommended to improve adaptability under varying traffic conditions. Furthermore, incorporating security-aware path selection and regularly updating the controller logic to reflect evolving network requirements would strengthen the practical relevance of the framework.

It is also advisable to validate the findings of this simulation-based study through testing on physical SDN hardware or hybrid testbeds, and to gather continuous performance data over extended periods to confirm the long-term benefits of SDN-based routing in real-world deployments.

CHAPTER 5 REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Academic Knowledge

Throughout the development of this project, I gained valuable academic knowledge in the areas of Computer Networks, Network Architecture, and Software-Defined Networking. This project enabled me to apply concepts learned from subjects such as Computer Networks, Data Communication, Network Protocols, and Distributed Systems. The practical implementation of the project helped bridge the gap between theoretical routing concepts and their real-world behaviour in a controlled, emulated environment.

I developed a deeper understanding of how routing protocols compute and maintain paths, how centralized control differs fundamentally from distributed control, and how network performance metrics such as throughput, latency, and packet loss reflect the underlying efficiency of a routing strategy. Working on a practical SDN evaluation project helped me understand how modern networks, particularly in cloud and data centre environments, are increasingly moving toward centralized, software-driven control.

5.2 Technical Skills

This project significantly improved my technical and networking skills. I gained hands-on experience configuring virtual network topologies using Mininet, writing controller logic using Python, and working with the OpenFlow protocol to manage flow tables on virtual switches. I learned how to capture and analyse network traffic using Wireshark and how to measure throughput and other performance metrics using tools such as iPerf.

I also enhanced my skills in scripting automated test scenarios, comparing protocol performance under controlled conditions, and interpreting network performance data to draw meaningful conclusions. These technical competencies will be highly beneficial for future work in network engineering, network automation, and SDN-based infrastructure design.

5.3 Problem-Solving and Critical Thinking

The project played an important role in developing my analytical thinking and problem-solving abilities. During development, I encountered challenges related to topology design, controller logic, and accurate performance measurement, all of which required careful planning, logical analysis, and systematic testing to resolve.

One key challenge involved designing test scenarios that could fairly and consistently compare routing strategies without introducing bias from the emulation environment itself. To address this, I evaluated multiple testing approaches, compared their trade-offs, and selected methods that produced the most reliable and repeatable results, which improved my decision-making skills and reinforced the importance of rigorous validation in technical projects.

5.4 Challenges Encountered and Overcome

Several technical and organisational challenges arose throughout the project lifecycle. Initially, understanding the complete behaviour of OpenFlow-based controllers and how flow rules propagate through a virtual topology required extensive research and experimentation.

Managing project timelines while coordinating between team members and balancing academic responsibilities was another significant challenge, which was addressed through effective scheduling and task division. Technical challenges such as debugging controller logic, resolving inconsistent flow table updates, and ensuring accurate measurement collection were addressed through repeated testing and troubleshooting, which enhanced my perseverance and confidence in handling complex networking tasks.

5.5 Application of Engineering Standards

The project allowed me to apply several networking and software engineering principles. I followed a modular development approach for controller logic to improve maintainability and readability. Standard IP addressing and topology documentation practices were followed to ensure consistency across all test scenarios.

Documentation standards were maintained throughout the project to record configurations, test parameters, and results for future reference. By following established engineering standards, I learned the importance of building reliable, reproducible, and well-documented technical systems that meet both academic and industry expectations.

5.6 Collaboration and Communication

Working as part of a team, I actively collaborated with my teammates and project guide to design the topology, divide implementation tasks, and review results together. These discussions helped improve both the technical quality of the project and our shared understanding of SDN concepts and routing behaviour.

Explaining technical concepts to teammates, presenting progress updates, and documenting our methodology improved my communication and presentation skills. I learned how to describe complex networking concepts clearly, which is an essential skill in professional and academic environments alike.

5.7 Insights into the Industry

This project provided valuable insight into the growing role of Software-Defined Networking in modern network infrastructure. I learned how cloud providers, data centres, and large enterprises increasingly rely on centralized, programmable control to manage complex, dynamic network environments efficiently.

The project highlighted the industry's ongoing shift toward automation, centralized visibility, and software-driven infrastructure management. Through this experience, I gained a better understanding of emerging trends in network engineering and developed skills that will help me adapt to future advancements in SDN, network automation, and cloud networking technologies.

CHAPTER 6 CONCLUSION

This project successfully evaluated the performance of routing protocols in a Software-Defined Networking environment, demonstrating the practical advantages of centralized, programmable network control over traditional distributed routing approaches. Using an emulated network topology, an OpenFlow-based SDN controller, and a suite of performance measurement tools, the project compared routing strategies under identical traffic and topology conditions to produce fair, reproducible results.

The implementation demonstrated how centralized routing decisions, driven by a global view of network state, can improve convergence speed, simplify traffic engineering, and adapt more effectively to changing network conditions than conventional distributed protocols. Performance monitoring confirmed measurable improvements in throughput and latency for SDN-based routing, particularly under scenarios involving topology changes or variable traffic load.

Throughout the project, valuable knowledge and practical experience were gained in network emulation, SDN controller programming, routing protocol design, and performance analysis. Technical challenges related to topology design, controller logic, and accurate measurement were successfully addressed through systematic planning, testing, and refinement.

In conclusion, this project confirms that Software-Defined Networking offers a reliable, flexible, and efficient approach to network routing, providing improved performance, easier management, and stronger adaptability to modern network demands. These findings support the continued adoption of SDN in enterprise, cloud, and data centre environments as networks continue to grow in scale and complexity.

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